# Sungmin Kang

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## Education

**Texas A&M University** College Station, TX, United States
Ph.D in Computer Science (Advisor: *Prof. Zhengzhong Tu*) Aug. 2026 - Present

**University of Southern California (USC)** Los Angeles, CA, United States
M.S. in Electrical Engineering (Advisor: *Prof. Salman Avestimehr*) Aug. 2024 - May. 2026
GPA: 4.00/4.00 *MS Honors Fellow*

**Sogang University** Seoul, Republic of Korea
B.S. in Electronic Engineering Mar. 2018 - Feb. 2024
GPA: 3.91/4.3 *Magna Cum Laude*

## Research Interests

Multimodal Generation, World Models, Agentic AI Systems, Trustworthy LLMs, Interpretability, Federated Learning

## Publications

C5. **Uncertainty as Feature Gaps: Epistemic Uncertainty Quantification of LLMs in Contextual Question-Answering**
Yavuz Faruk Bakman, Sungmin Kang, Zhiqi Huang, Duygu Nur Yaldiz, Catarina G Belém, Chenyang Zhu, Anoop Kumar, Alfy Samuel, Daben Liu, Salman Avestimehr, Sai Praneeth Karimireddy
*The Fourteenth International Conference on Learning Representations* (*ICLR 2026*) [PDF]

C4. **GEM: A Scale- and Distribution-Aware Sparse Fine-Tuning Framework for Effective Downstream Adaptation**
Sungmin Kang, Jisoo Kim, Salman Avestimehr, Sunwoo Lee
*The 40th Annual AAAI Conference on Artificial Intelligence* (*AAAI 2026*) [PDF]

C3. **Layer-wise Update Aggregation with Recycling for Communication-Efficient Federated Learning**
Jisoo Kim, Sungmin Kang, Sunwoo Lee
*The Thirty-Ninth Annual Conference on Neural Information Processing Systems* (*NeurIPS 2025*) [PDF]

C2. **TruthTorchLM: A Comprehensive Library for Predicting Truthfulness in LLM Outputs**
Duygu Nur Yaldiz*, Yavuz Faruk Bakman*, Sungmin Kang, Alperen Ozis, Hayrettin Eren Yildiz, Mitash Shah, Zhiqi Huang, Anoop Kumar, Alfy Samuel, Daben Liu, Sai Praneeth Karimireddy, Salman Avestimehr
*The 2025 Conference on Empirical Methods in Natural Language Processing* (*EMNLP 2025*) [PDF] [Github]

C1. **Reconsidering LLM Uncertainty Estimation Methods in the Wild**
Yavuz Faruk Bakman*, Duygu Nur Yaldiz*, Sungmin Kang, Tuo Zhang, Baturalp Buyukates, Salman Avestimehr, Sai Praneeth Karimireddy
*63rd Annual Meeting of the Association for Computational Linguistics* (*ACL 2025*) [PDF]

M1. **Uncertainty Quantification for Hallucination Detection in Large Language Models: Foundations, Methodology, and Future Directions**
Sungmin Kang, Yavuz Faruk Bakman, Duygu Nur Yaldiz, Baturalp Buyukates, Salman Avestimehr
*IEEE BITS the Information Theory Magazine, 2025* [PDF]

## Research Experiences

**Graduate Research Intern,** Argonne National Laboratory Jun. 2026 - Aug. 2026
Host: *Dr. Kibaek Kim*, Mathematics and Computer Science Division

- **Federated Learning Simulator**: Built a simulator that faithfully reproduces realistic FL environments—from large-scale cross-device settings to high-performance computing—and reports system cost (compute, communication, queueing)
- **Agentic Paper-to-Code for FL**: Developed an LLM-agent pipeline that reads an FL research paper and automatically implements it on existing frameworks (APPFL, Flower, NVFlare), producing framework-compliant code verified by an automated rubric

**Graduate Research Assistant,** University of Southern California Oct. 2024 - Present
Advisor: *Prof. Salman Avestimehr*, Information Theory and Machine Learning (vITAL) Lab

- **LLM Uncertainty Quantification** [C1, C5]: Proposed a UQ method that measures a model's feature gap in its representation space using contrastive prompts; empirically evaluated existing UQ methods under realistic deployment conditions

such as calibration, robustness to input variation, and ensembling
- **Open-Source & Survey** [C2, M1]: Built a truthfulness-evaluation library (25+ methods) and authored a survey that categorizes and organizes 35 representative UQ methods

**Graduate Research Intern,** Inha University — May. 2024 - Dec. 2025
with *Prof. Sunwoo Lee*, Large-Scale Machine Learning Systems Lab

- **Communication-Efficient Federated Learning** [C3]: Designed an FL algorithm that selects and recycles layer updates to reduce communication cost
- **Parameter-Efficient Fine-Tuning** [C4]: Proposed a sparse fine-tuning framework that uses the gradient-to-weight ratio to identify and tune only the important parameters for effective downstream adaptation

**Undergraduate Research Assistant,** Sogang University — Jul. 2023 - Apr. 2024
Advisor: *Prof. Hongseok Kim*, Networking for Intelligence Computing and Energy (NICE) Lab

- **Federated Learning Systems**: Built a real wireless FL system on 8 laptops with a multi-threaded server communicating over sockets [Github]
- **Distributed Optimization for FL**: Designed an FL algorithm that predicts client updates based on a distributed-optimization formulation for accelerated convergence

**Undergraduate Research Intern,** Seoul National University — Jan. 2023 - Apr. 2023
with *Prof. Taesup Moon*, Machine Intelligence and Data science (M.IN.D) Lab

- **Multimodal Medical Imaging**: MRI classification for predicting Alzheimer's disease progression via transfer learning, addressing limited data and class imbalance

## Projects

**TruthTorchLM: Open-Source Hallucination Detection Library**, USC & Capital One — Oct. 2024 - May 2025

- Developed an open-source library with 25+ methods for evaluating LLM truthfulness [Github] [PDF]
- Provides a unified interface to calibrate, evaluate UQ methods, supporting long-form and contextual QA for both API and HuggingFace models

## Honors and Awards

- MS Honors Fellowship, USC — May 2026
- Outstanding Academic Achievement Award (one master's student department-wide), USC — May 2026
- Student Travel Scholarship & Volunteer, AAAI-26 — Jan. 2026
- Best Poster Award (out of 110+ teams), ECE 15th Annual Research Festival, USC — Oct. 2025
- Best Poster Award (out of 50+ teams), Viterbi PhD Visit Day, USC — Apr. 2025
- 3rd Prize (out of 50+ teams), Senior Thesis Project Contest, Sogang University — Dec. 2023
- 3rd Prize (out of 40+ teams), Capstone Design Project Contest, Sogang University — Jun. 2023
- Daesang Foundation Scholarship (Merit-based scholarship, 5M KRW/semester) — Spring 2019 - Fall 2023

## Teaching Experiences

- Grader, EE 503: Probability for Engineers by *Prof. Kosko Bart*, USC — Fall 2025
- Teaching Assistant, EEE 4171: AI Communications by *Prof. Hongseok Kim*, Sogang University — Spring 2024
- Teaching Assistant, COR 1010: AI Programming by *Prof. Naeun Jang*, Sogang University — Summer 2023
- Education Volunteer Services for financially disadvantaged teenage students — 2019, 2021

## Academic Services

- *Invited Talk:* "Federated Learning Implementation through Socket Communication", Sogang University — May 2024
- *Reviewer:* IEEE Transactions on Artificial Intelligence